MIXED COMPOSITIONS

# The triangle method

It converts the problem from inverse trigonometric territory back to right triangle ratios.

01

MIXED COMPOSITION — BEFORE

$$\sin(\arccos x)$$

Name the angle — let θ = arccos x

02

ONE ANGLE, ONE RATIO

$$\cos\theta = \frac{x}{1}$$

Build the triangle — adjacent x, hypotenuse 1, so the opposite side is $\sqrt{(1 - x^2)}$; sine is opposite over hypotenuse

03

ALGEBRAIC FORM — AFTER

$$\sin(\arccos x) = \sqrt{1 - x^2}$$

▼ WHERE IT HOLDS